

DAY 07 – IP CONFIGURATION

Networking Assignment Report

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Assignment Code	DAY07-IP-CONFIG
Total Marks	20
Operating System	Ubuntu Linux

Ubuntu Linux Note: The assignment specifies the Windows command ipconfig /all. This practical was performed on Ubuntu Linux, so the Linux equivalents ip route and ifconfig -a were used instead.

THEORY (8 MARKS)

1. What is a network address?

A network address identifies the network/subnet in entirety. It is the first address in a subnet which marks the network, not a specific host.

2. What is a broadcast address?

A broadcast address is the last IPv4 subnet address; it is used for transmitting data packets to all nodes and hosts in a subnet.

3. What is a usable host address?

A usable host address is an IP address which can be assigned to a computer, a router interface, or the like. The usable address comes between the network and broadcast address.

4. What is the first usable IP address?

The first usable IP address is the IP address right after the network address.

Example for 192.168.1.0/24: 192.168.1.1

5. What is the last usable IP address?

The last usable IP address is the IP address right before the broadcast address.

For 192.168.1.0/24: 192.168.1.254

6. What does /24 mean?

/24 is a CIDR notation which defines that 24 bits out of an IPv4 address are assigned to the network portion; 8 bits remain to represent the host part.

7. What subnet mask corresponds to /24?

The subnet mask corresponding to /24: 255.255.255.0

8. What is a default gateway?

A default gateway is a router's IP address which an IP device utilizes to communicate with networks outside its domain.

9. Why does a router interface need an IP address?

A router interface needs an IP address to be able to identify itself in the target network as part of that network and to carry out Layer 3 functions. It allows the router to make decisions about routing.

10. What is a static IP configuration?

A static IP configuration refers to a manual method of setting an IP address, subnet mask, and default gateway on a device, as compared to dynamic IP allocation.

11. What does “show ip interface brief” display?

In general, show ip interface brief on Cisco devices displays a brief summary of router interfaces, usually with the following information:

- Interface name,
- IP address,
- Interface status,
- Protocol status.

It is mostly used to examine the interfaces to evaluate their status.

12. What does “no shutdown” do conceptually on a Cisco interface?

no shutdown command on a Cisco interface opens it to traffic; conceptually it allows it to operate.

13. What does interface status “up/up” generally indicate?

In general, the interface status “up/up” means:

- First up – the physical interface/link is up,
- Second up – the line protocol is up.

So, in brief, the interface is operating fine with no issues detected.

14. Why should network configuration always be verified?

The network configuration should be verified to ensure that the address, subnet mask, gateway, interface status, and routing are properly set. It will prevent configuration issues causing communication problems.

15. Why is IPv4 configuration important in cybersecurity?

IPv4 configuration is important in cybersecurity because IP addresses are crucial for identification and communication of devices; knowledge about IPv4 is fundamental in determining and analyzing any network-related events in cybersecurity.

IPv4 EXERCISE (4 MARKS)

For the network 192.168.100.0/24:

Item	Answer
Network Address	192.168.100.0
First Usable IP	192.168.100.1

Last Usable IP	192.168.100.254
Broadcast Address	192.168.100.255

PRACTICAL (4 MARKS)

The assignment requests `ipconfig /all`. Since this practical was performed on Ubuntu Linux, the equivalent commands `ifconfig -a` and `ip route` were used to record the same information.

Field	Value
IPv4 Address	172.20.10.3
Subnet Mask	255.255.255.240
Default Gateway	172.20.10.1
DNS Server	172.20.10.1

SCREENSHOTS (2 MARKS)

```

md-hamza-khan@md-hamza-khan-HP-ZBook: ~
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ifconfig -a
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether e8:d8:d1:ec:91:9e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xe8100000-e8120000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 5115 bytes 725601 (725.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5115 bytes 725601 (725.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp0s20f3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.20.10.3 netmask 255.255.255.240 broadcast 172.20.10.15
    inet6 2409:40e3:5001:3fa2:f646:5963:607c:ce11 prefixlen 64 scopeid 0x0<global>
    inet6 2409:40e3:5001:3fa2:2ed7:722c:9b7d:7dac prefixlen 64 scopeid 0x0<global>
    inet6 fe80::be09:a807:591a:d68b prefixlen 64 scopeid 0x20<link>
    ether c8:09:a8:e6:1e:f5 txqueuelen 1000 (Ethernet)
    RX packets 115076 bytes 11482000 (114.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 52327 bytes 31959933 (31.9 MB)
    TX errors 0 dropped 31 overruns 0 carrier 0 collisions 0

wlx00e02691d31b: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 00:e0:26:91:d3:1b txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
  
```

Figure 1 – `ifconfig -a` (IPv4 address, subnet mask, adapter)

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip route
default via 172.20.10.1 dev wlp0s20f3 proto dhcp src 172.20.10.3 metric 600
172.20.10.0/28 dev wlp0s20f3 proto kernel scope link src 172.20.10.3 metric 600
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.059 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.052 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3085ms
rtt min/avg/max/mdev = 0.043/0.052/0.059/0.006 ms
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ping -c 4 172.20.10.1
PING 172.20.10.1 (172.20.10.1) 56(84) bytes of data.
64 bytes from 172.20.10.1: icmp_seq=1 ttl=64 time=4.35 ms
64 bytes from 172.20.10.1: icmp_seq=2 ttl=64 time=3.86 ms
64 bytes from 172.20.10.1: icmp_seq=3 ttl=64 time=4.84 ms
64 bytes from 172.20.10.1: icmp_seq=4 ttl=64 time=4.45 ms

--- 172.20.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 3.861/4.376/4.843/0.350 ms
md-hamza-khan@md-hamza-khan-HP-ZBook:~$
```

Figure 2 – ip route, ping 127.0.0.1, ping default gateway (172.20.10.1)

COMMANDS USED

ifconfig -a

ip route

ping -c 4 127.0.0.1

ping -c 4 172.20.10.1

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Platform: Ubuntu Linux